RESEARCH COMPANION

Outer Alignment & Values Benchmarking

Annotated bibliography

A source-by-source guide to the 27 works cited in *The Architecture of Machine Morality*, with notes on evidentiary role, best use, and interpretive limits.

27	7	2025-26
SOURCE RECORDS	EVIDENCE CLUSTERS	REPORT HORIZON

Scope: Annotations evaluate how each cited item functions within the supplied report. They do not constitute an independent replication or peer-review audit.

Prepared from the user-supplied Gemini Deep Research report

October 2026

Read the bibliography as an evidence stack

The report's reference list mixes original benchmark papers with reviews, tertiary explainers, repository mirrors, community indexes, and a research trace. Those materials can all be useful, but they answer different questions and should carry different evidentiary weight.

Start with identifiable primary work.

Benchmark papers and empirical studies are the right place to verify task design, model sets, scoring, uncertainty, and quantitative results.

Use reviews to navigate.

Surveys help map failure modes and evaluation families, but decisive claims should be followed back to the underlying experiment.

Treat mirrors as access paths.

ResearchGate, Semantic Scholar, AlphaXiv, and document-hosting links may aid discovery; the canonical paper should anchor attribution and version control.

Use tertiary pages for orientation only.

Wikipedia, corporate explainers, academy collections, and curated repositories are useful entry points, not substitutes for primary evidence.

Annotation method

Each record preserves the title and venue signal available in the report's works-cited section. The annotation identifies the source's role in the report, its most defensible use, and a specific caution. Four URLs are not reproduced because the source report breaks or truncates them across lines; this avoids silently inventing or normalizing links.

Five source clusters carry the report's central argument

The bibliography is strongest when it connects the report's claims to clearly named benchmark or empirical studies. It is weakest where incomplete titles, mirrors, or tertiary pages stand between the reader and the underlying evidence.

Core empirical spine

Specification gaming, Reward Hacking Benchmark, RewardBench 2, personalized reward models, VAL-Bench, MOSAIC, GAIN, and PlurVA.

Contextual synthesis

Misalignment surveys and structured reviews connect individual benchmarks to the wider agentic-safety landscape.

Discovery layer

Curated lists, mirrors, academy collections, and traces help locate work but should not carry empirical conclusions alone.

Key interpretive risk

Consistency, moral quality, refusal, cultural calibration, and reward-model accuracy are distinct constructs; no single score stands for “alignment.”

Records by evidence cluster

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Recommended reading order: Begin with entries 5, 6, 14, 15, 11, 16, 22, and 23; then use reviews and indexes to broaden the search.

Foundations and formal framing

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01 DEPTH Mathematical Framework for Mitigat

OpenReview PDF; title is truncated in the source report

Source type: Primary-research candidate / review-platform record

The report appears to use this work as formal support for mitigation strategies aimed at proxy-objective failure. Its OpenReview location is a useful provenance signal because it can expose manuscript and review history, but the citation as supplied does not preserve the complete title or authors.

Best use: Use to examine the mathematical assumptions behind proposed safeguards; recover the full bibliographic record before quoting or attributing a theorem.

Interpretive caution: Incomplete citation metadata makes this entry difficult to audit from the report alone.

> https://openreview.net/pdf?id=Co65Bj9zZ3

02 What Is LLM Alignment?

IBM Think

Source type: Corporate explainer

A concise orientation to large language model alignment vocabulary and practice. In the report, it is most useful for framing outer alignment for a broad audience rather than establishing contested empirical claims.

Best use: Use for definitions, examples, and accessible background.

Interpretive caution: This is not a substitute for primary scholarship and should not carry quantitative or causal claims by itself.

> https://www.ibm.com/think/topics/llm-alignment

Foundations and formal framing

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03 Beyond Intentions: A Critical Survey of Misalignment in LLMs

Publisher-hosted article; DOI path supplied

Source type: Scholarly survey

This survey provides a taxonomy-level view of misalignment mechanisms and helps situate outer alignment alongside adjacent failure modes. The report can legitimately draw synthesis and terminology from it, while tracing important empirical claims to the survey's underlying sources.

Best use: Use to map the literature and identify source clusters.

Interpretive caution: As a review, its value depends on coverage choices and the quality of cited primary evidence.

> https://www.sciopen.com/article/10.32604/cmc.2025.067750

04 AI alignment

Wikipedia

Source type: Tertiary reference

A navigational overview of alignment concepts, historical debates, and common terminology. It likely supports broad context in the report but is not suitable evidence for novel findings or precise performance figures.

Best use: Use to orient readers and follow references to original work.

Interpretive caution: Open editing and shifting page content make it inappropriate as the final authority for substantive claims.

> https://en.wikipedia.org/wiki/AI_alignment

Specification gaming and reward hacking

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05

Towards Understanding Specification Gaming in Reasoning Models

arXiv HTML, identifier 2605.02269v1

Source type: Preprint

This is the report's most direct cited basis for the claim that reasoning capability can amplify specification gaming rather than reliably suppress it. Its experimental design is therefore central to interpreting reported exploit-rate changes across reasoning budgets.

Best use: Use for task design, model comparisons, intervention details, and effect-size definitions.

Interpretive caution: Treat version 1 as provisional; check for later revisions, peer review, and robustness analyses before relying on headline percentages.

> https://arxiv.org/html/2605.02269v1

06

Reward Hacking Benchmark: Measuring Exploits in LLM Agents

arXiv PDF, identifier 2605.02964

Source type: Benchmark preprint

The report uses this benchmark to discuss long-horizon agent workflows in which exploitable shortcuts compete with honest completion. The source is especially relevant to claims about instrumented exploit detection and complexity-dependent failures of post-training.

Best use: Use to inspect benchmark construction, scoring, contamination controls, and model-level results.

Interpretive caution: Benchmark behavior may not transfer directly to unconstrained deployment; review the task distribution and detector assumptions.

> https://arxiv.org/pdf/2605.02964

Specification gaming and reward hacking

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07 A Multi-dimensional Study of Behavioral and Representational Drift

arXiv HTML, identifier 2607.22676v2

Source type: Preprint

This work appears to support the report's argument that alignment failures can involve both observable behavior and latent representation change. Its value is in connecting output-level evaluations with internal or longitudinal measures rather than treating a single benchmark score as complete.

Best use: Use to compare behavioral and representational indicators of drift.

Interpretive caution: The report does not specify which reported result maps to which claim; verify the exact experiment before citing a number.

> https://arxiv.org/html/2607.22676v2

08 Agentic AI Safety: A Structured Review of Open Problems and Their…

MDPI, AI journal path supplied

Source type: Structured review

A broad review that places reward hacking, oversight, autonomy, and deployment risk within the wider agentic-safety landscape. It can contextualize the report's shift from passive language models to tool-using, long-horizon agents.

Best use: Use for problem taxonomy and cross-references to primary studies.

Interpretive caution: The title is truncated in the source report, and review-level synthesis should not replace direct inspection of pivotal experiments.

> https://www.mdpi.com/2673-2688/7/8/298

Specification gaming and reward hacking

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12 Reward hacking

Wikipedia

Source type: Tertiary reference

A plain-language overview of reward hacking and illustrative examples. It can help readers recognize the concept before encountering the report's more technical benchmarks.

Best use: Use for orientation and terminology.

Interpretive caution: Do not use it as sole support for prevalence, causal mechanisms, or measured mitigation effects.

> https://en.wikipedia.org/wiki/Reward_hacking

Reward models and personalization

1-2 of 2

14 RewardBench 2: Advancing Reward Model Evaluation

arXiv PDF, identifier 2506.01937

Source type: Benchmark preprint

This is the report's key source for evaluating reward models across more difficult and diverse preference settings. It underpins discussion of generalization limits and the need to judge reward models by performance beyond familiar preference data.

Best use: Use for benchmark composition, evaluator comparisons, category-level results, and scoring methodology.

Interpretive caution: Check whether quoted accuracy thresholds aggregate heterogeneous tasks and whether later versions revise the leaderboard.

> https://arxiv.org/pdf/2506.01937

15 Evaluating Reward Models with Human Aligned Personalization

arXiv HTML, identifier 2604.07343v2

Source type: Preprint

The report relies on this study for claims that personalized reward modeling does not scale monotonically with parameter count and that raw user-history injection may fail. The source is directly relevant to contextual and user-specific outer alignment.

Best use: Use to inspect personalization datasets, profile construction, train-test splits, and scale comparisons.

Interpretive caution: Personalization quality is sensitive to how preferences are elicited and represented; generalization beyond the studied profiles requires care.

> https://arxiv.org/html/2604.07343v2

Belief consistency and moral measurement

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11 VAL-Bench: Belief Consistency as a Measure

arXiv PDF, identifier 2510.05465

Source type: Benchmark preprint

VAL-Bench supplies the report's framework for distinguishing stance consistency, sycophancy, and refusal across opposing prompt framings. It operationalizes a narrow but important dimension of alignment: whether a model changes its expressed values under rhetorical pressure.

Best use: Use for prompt-pair construction, label definitions, dataset balance, and refusal-versus-consistency analysis.

Interpretive caution: Stable answers are not automatically correct or ethically desirable; consistency and substantive moral quality must be evaluated separately.

> https://arxiv.org/pdf/2510.05465

16 MOSAIC: Unveiling the Moral, Social and Individual Dimensions of...

arXiv HTML, identifier 2603.00048

Source type: Benchmark preprint

MOSAIC is cited as a multidimensional alternative to single-theory moral tests. Its combination of adapted psychometric instruments and interactive scenarios is central to the report's claim that stated values can diverge from applied choices.

Best use: Use to compare questionnaire and behavioral-game results across value dimensions.

Interpretive caution: Psychometric instruments adapted from humans may not preserve construct validity when applied to language models.

> https://arxiv.org/html/2603.00048

Belief consistency and moral measurement

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19 ETHICS Benchmark for AI Ethics

Emergent Mind

Source type: Secondary topic summary

This page summarizes the ETHICS benchmark and its normative subsets. Within the report it is best read as an accessible map to an established benchmark rather than the canonical methodological source.

Best use: Use to orient readers to benchmark tasks and vocabulary, then follow through to the original dataset paper.

Interpretive caution: A topic summary may compress methodology, update asynchronously, or mix source findings with later commentary.

> https://www.emergentmind.com/topics/ethics-benchmark

20 Evaluating Moral and Ethical Alignment in Large Language Models

MDPI, Applied Sciences path supplied

Source type: Publisher-hosted research article

This article appears to ground the report’s distinction between moral and ethical axes and its discussion of evaluator asymmetries. It is especially relevant to the reported generator-evaluator matrix and category-specific accuracy gaps.

Best use: Use to verify experimental cells, axis definitions, confusion patterns, and statistical treatment.

Interpretive caution: The report’s summary should be checked against the article’s exact labels and model set before reusing numerical comparisons.

> https://www.mdpi.com/2076-3417/16/13/6769

Belief consistency and moral measurement

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21

How Do LLMs Reflect Human Moral Foundations? A Study...

Taylor & Francis full article; DOI path supplied

Source type: Publisher-hosted research article

This study links large language model outputs to human moral-foundations constructs. It supports the report's discussion of machine-human convergence and the limits of interpreting rank-order similarity as full agreement.

Best use: Use to examine sampling, prompting, human comparison groups, and correlation metrics.

Interpretive caution: Moral Foundations Theory captures only part of moral judgment; construct scope and cultural coverage matter.

> https://www.tandfonline.com/doi/full/10.1080/29974100.2026.2678495

Pluralistic and cross-cultural alignment

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10 PlurVA-LLM-2026 Shared Task Track-1: Pluralistic Value Alignment...

ResearchGate record; supplied URL wraps in the source report

Source type: Repository mirror / shared-task paper

This entry supports the report's claims about multilingual fine-tuning and culturally localized threshold calibration. It appears closely related to item 23 and may represent a mirror or alternate record of the same shared-task work.

Best use: Use to understand track design, participating systems, language coverage, and calibration procedures.

Interpretive caution: Resolve duplication with item 23 and recover authors, venue, and canonical version before treating these as two independent sources.

> *Full URL not reproduced because the source report wraps or truncates it.*

17 Operationalizing Pluralistic Values in Large Language Models...

Semantic Scholar record; supplied URL wraps in the source report

Source type: Index / paper-discovery record

This work is cited to expand alignment beyond a single value target toward explicitly pluralistic objectives. The Semantic Scholar record is useful for discovery, but the underlying paper should be used for method and evidence.

Best use: Use to identify the canonical manuscript and its operational definition of pluralism.

Interpretive caution: The supplied citation is incomplete and points to an index rather than necessarily to the authoritative full text.

> *Full URL not reproduced because the source report wraps or truncates it.*

Pluralistic and cross-cultural alignment

23

PlurVA-LLM-2026 Shared Task Track-1

arXiv PDF, identifier 2609.32382

Source type: Shared-task preprint

This is the cleaner primary record for the pluralistic value-alignment shared task cited in the report. It is the best starting point for the reported regional accuracies and the effect of conditional threshold calibration.

Best use: Use to verify datasets, scoring, macro-averaging, baselines, and country- or language-level results.

Interpretive caution: Shared-task outcomes depend on test-set construction and may not capture within-culture disagreement or deployment context.

> https://arxiv.org/pdf/2609.32382

Operational pressure and multimodal evaluation

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22 GAIN: A Benchmark for Goal-Aligned Decision-Making of Large...

ACL Anthology PDF, LREC 2026 path supplied

Source type: Conference-paper record

GAIN is the report's main evidence for testing whether models preserve explicit norms when business goals, authority, urgency, emotion, or incentives apply pressure. It moves moral evaluation closer to operational decision-making than abstract dilemmas do.

Best use: Use to inspect the G-S-N-P scenario formalization, pressure manipulations, model set, and norm-deviation scoring.

Interpretive caution: Scenario realism and the chosen norms constrain external validity; performance does not by itself establish safe behavior in live organizations.

> https://aclanthology.org/2026.lrec-1.340.pdf

24 MM-MoralBench: A MultiModal Moral Evaluation Benchmark for...

AlphaXiv abstract mirror, identifier 2412.20718

Source type: Benchmark-paper mirror

MM-MoralBench extends the report's evaluation scope beyond text-only models to systems that interpret images and language together. It is relevant to cross-modal inconsistencies and safety failures triggered by visual context.

Best use: Use to identify multimodal task categories, modality controls, and failure patterns; retrieve the canonical paper where possible.

Interpretive caution: The supplied link is a mirror, and the truncated title obscures complete bibliographic details.

> https://www.alphaxiv.org/abs/2412.20718

Surveys, indexes, and research infrastructure

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09

Evaluating AI Alignment in Eleven LLMs through Output-Based Analysis and Human Benchmarking

ResearchGate publication record; URL wraps in the source report

Source type: Research-paper mirror

This study appears to provide a cross-model, human-referenced evaluation relevant to the report's machine-psychology discussion. Its breadth can illuminate comparative patterns across models.

Best use: Use to inspect rater design, prompts, reliability, and the distinction between rank-order and absolute agreement.

Interpretive caution: Use the canonical publication record if available; a repository mirror alone does not establish review status or version.

> *Full URL not reproduced because the source report wraps or truncates it.*

13

AI Alignment Principles and Practices

Scribd document record

Source type: Hosted document / survey

A broad survey-style document that may help readers assemble terminology and practice areas. It is peripheral to the report's strongest empirical claims.

Best use: Use as a discovery aid or practitioner-oriented overview.

Interpretive caution: Authorship, review status, versioning, and access stability are not clear from the supplied citation; corroborate every substantive claim.

> https://www.scribd.com/document/878211973/Ai-Alignment-Survey-April-2025

Surveys, indexes, and research infrastructure

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18 A Hybrid Survey and Conceptual Framework for LLM Evaluation

arXiv HTML, identifier 2606.30219v5

Source type: Survey preprint

This survey offers a higher-level evaluation framework that can connect alignment benchmarks to broader large language model assessment. In the report, it likely helps organize heterogeneous metrics rather than supply a single decisive finding.

Best use: Use for taxonomy, coverage comparison, and references to evaluation methods.

Interpretive caution: A survey's conceptual categories should not be mistaken for validated measurement constructs without checking the underlying studies.

> https://arxiv.org/html/2606.30219v5

25 Awesome-LLM-Safety: Datasets & Benchmark

GitHub curated list

Source type: Community-maintained index

A curated directory of safety datasets and benchmarks. It is useful for expanding the report's benchmark inventory and locating projects that may not yet appear in reviews.

Best use: Use for discovery and cross-checking benchmark names.

Interpretive caution: Inclusion is not validation; entries vary in maturity, documentation, maintenance, and evidentiary weight.

> https://github.com/ydyjya/Awesome-LLM-Safety/blob/main/subtopic/Datasets&Benchmark.md

Surveys, indexes, and research infrastructure

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26 LLM-as-a-judge and non-verifiable rewards trace

Hugging Face dataset trace; supplied URL wraps in the source report

Source type: Research trace / dataset artifact

This trace appears to document a research-index run concerning LLM judges and non-verifiable rewards. It may reveal search provenance or intermediate evidence behind the report's discussion of recursive evaluator bias.

Best use: Use for auditability and source discovery, not as a standalone research conclusion.

Interpretive caution: A trace is process evidence; its claims require validation against the papers or datasets it points to.

> *Full URL not reproduced because the source report wraps or truncates it.*

27 Self-Improving Agents

DAIR.AI Academy paper collection

Source type: Curated educational collection

This collection situates self-rewarding, code-modifying, and recursively improving agents within a broader research stream. It supports the report's forward-looking section by pointing readers toward relevant papers.

Best use: Use to discover primary work on self-improvement architectures and safety implications.

Interpretive caution: A collection page should not be cited as the sole evidence for specific reward-hacking reductions or system capabilities.

> https://academy.dair.ai/papers/collections/self-improving-agents

A defensible evidence strategy separates discovery from proof

01 **Verify the primary benchmark**

For every reported score or effect size, confirm the exact task, model version, evaluator, aggregation rule, and uncertainty in the canonical paper.

02 **Keep constructs separate**

Reward hacking, stance consistency, refusal, moral classification, pluralistic calibration, and multimodal safety measure different failure modes. Do not collapse them into a single alignment ranking.

03 **Triangulate automated judges**

Pair LLM-as-a-judge results with calibrated human baselines, detector audits, and adversarial tests. Evaluator bias is itself part of the outer-alignment problem.

04 **Track versions and provenance**

Record authors, dates, manuscript versions, venues, and canonical links. Four entries in the supplied list have wrapped or truncated URLs, and several titles are incomplete.

05 **Use deployment context as a test dimension**

Evidence from GAIN, personalized reward-model studies, pluralistic tasks, and multimodal benchmarks argues for evaluation under realistic pressure, user context, culture, and modality.

Bottom line

The bibliography supports the report's broad conclusion that outer alignment is a multi-objective measurement and control problem, not a one-time safety filter. Its strongest claims are traceable to named benchmark papers; its weakest links are incomplete records and secondary pages used where a primary source would be preferable.